

Department of Mathematics and Physics

Course Information

Course Name	Calculus I	**Instructor**	Md Ahsanul Haque, Ph.D.
Course Code	MAT-120	**Office Room**	SAC 1133
Section No	11	**Office Hours**	ST 1:00-4:15 PM, MW 2:35-4:15 PM
Semester	Summer 2026	**Email**	ahsanul.haque01@northsouth.edu
Class Time	MW 06:00 PM-07:30 PM	**Location**	NAC 314
Credit Hours	3:0	**Prerequisites**	MAT 116

Course Description

This course is intended to develop practical skills in differential and integral calculus including their applications to various technical problems. The basic differential rules will be introduced as well as the methods of differentiating algebraic and transcendental functions will be developed. The definite integrals and indefinite integrals along with its applications in finding the area will be studied.

Course Objectives

- To classify different types of functions, approximate their limits both numerically and graphically as well as their continuity at a point or determine the intervals of continuity.
- To apply the concept of limits and continuity to understand the principles of differentiation and integration.
- To differentiate all types of functions including implicit and explicit type using different methods; apply differentiation to determine the rate of change and maxima and minima of functions.
- To integrate different types of functions either by using the integral table or substitution technique; apply the definite integral to obtain the area under curves.

Student Learning Outcomes (COs)

Upon the successful completion of this course, a student will be able to:

- **CO-1:** Demonstrate the ability to identify the type of a given function, approximate its limit both numerically and graphically as well as its continuity at a given point or determine the intervals of continuity of the function.
- **CO-2:** Determine the differentiability and integrability of functions using the concepts of limits and continuity.
- **CO-3:** Differentiate various types of functions comprising both the implicit and explicit types using different methods and apply differentiation to determine the rate of change and maxima and minima of functions.

- **CO-4:** Evaluate the Integrals associated with different types of functions either by using the integral table or substitution technique; interpret the geometric meaning of integral and apply this concept to solve geometric and physical problems, such as finding the area under curves.

Mapping of Course Outcomes

CO	Description (Abbreviated)	Bloom's	Delivery	Assessment
CO-1	Limits and Continuity	C1, C2, C3	Lectures, Notes	Quiz, Assignment
CO-2	Differentiability and Integrability	C3, C4	Lecture, Discussion	Midterms, Participation
CO-3	Differentiation Methods and Apps	C1, C3, P2	Lecture, Discussion	Midterms, Assignment
CO-4	Integration and Area	C1, C3, P2	Lecture, Discussion	Final Exam, Assignment

Learning Resources

Textbook	Howard Anton, *Calculus, Early Transcendentals*, 10th Edition, 2013
Reference	James Stewart, *Calculus*, 7th Edition, 2012

Assessment Strategy and Grading Policy

Grading Tools

Attendance	05%
Assignments	10%
Quizzes	20%
Midterm	30%
Final Exam	35%

Grading Scale

Numerical Scores	Letter Grade	Grade Points
93 +	A	4.0
90 - 92	A-	3.7
87 - 89	B+	3.3
83 - 86	B	3.0
80 - 82	B-	2.7
77 - 79	C+	2.3
73 - 76	C	2.0
70 - 72	C-	1.7
67 - 69	D+	1.3
60 - 66	D	1.0
Below 60	F	0.0

Policies

- **Classroom Rules:** Electronic devices (cell phones, laptops, etc.) are strictly prohibited. Absolute discipline is imperative. Arrive on time.
- **Academic Integrity:** Department does not tolerate academic dishonesty (cheating, fabrications, etc.). Violations are treated seriously.
- **Exams & Makeup:** Best two out of three quizzes. No makeup quizzes or midterms under any circumstances.
- **Attendance:** Students missing 25% or more of classes will fail (NSU mandate).

Rights of the instructor

The instructor reserves the right to change a policy as needed. If any such change is made, the instructor will announce it in class as well as in Canvas.

Course Contents & Tentative Lecture Plan

Lec	Topic	Activities	Assess.	Outcome	Art.
1	Introduction, Functions, Families of Functions	Discussions	Quiz 1	CO-1	0.1-0.3
2	Inverse Functions, Inverse Trigonometric Functions	Discussions	Quiz 1	CO-1	0.4
3	Exponential and Logarithmic Functions	Discussions	Quiz 1	CO-1	0.5
4	Limits, Computing Limits, End Behavior	Quiz 1, Midterm	Assign I	CO-1	1.1-1.3
5	Continuity, Continuity of Functions	Midterm	Assign I	CO-1	1.5, 1.6
6	Tangent lines, Rates of Change, Derivative Function	Midterm	Assign I	CO-2,3	2.1, 2.2
7	Techniques of Differentiation, Product/Quotient Rules	Midterm	Assign I	CO-3	2.3, 2.4
8	Derivatives of Trig Functions, Chain Rule	Midterm	Assign I	CO-3	2.5, 2.6
9	Implicit Differentiation	Midterm	Assign I	CO-3	3.1
10	Derivatives of Logarithmic Functions	Midterm	Assign I	CO-3	3.2
11	**Midterm Exam**				
12	Derivatives of Exp and Inv Trig Functions	Quiz 2	Assign II	CO-3	3.3
13	Related Rates	Quiz 2	Assign II	CO-3	3.4
14	L'Hopital's Rule; Indeterminate Forms	Quiz 2	Assign II	CO-2	3.6
15	L'Hopital's Rule; Indeterminate Forms	Quiz 2, Final	Assign II	CO-2	3.6
16	Analysis of Functions: Increase, Decrease, Concavity	Quiz 3, Final	Assign III	CO-3	4.1
17	Relative Extrema, Absolute Maxima and Minima	Quiz 3, Final	Assign III	CO-3	4.2, 4.4
18	The Indefinite Integral	Quiz 3, Final	Assign III	CO-4	5.2
19	Integration by Substitution	Quiz 3, Final	Assign III	CO-4	5.3
20	Area as a Limit; Sigma Notation	Final Exam	Assign IV	CO-4	5.4
21	The Definite Integral	Final Exam	Assign IV	CO-4	5.5
22	The Fundamental Theorem of Calculus	Discussions	-	CO-4	5.6
23	Evaluating Definite Integrals by Substitution	Final Exam	Assign IV	CO-4	5.9
24	Review and discussion about final exam	-	-	-	-
Final Exam (Declared by Controller of Examinations)					